



ADITYA WAHYU SETIAWAN

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Undergraduate student from the Electrical Engineering department at State University of Malang. I have a strong interest in power systems, including power generation, distribution, and management, as well as renewable energy integration. I also have expertise in the design, development, and repair of electronic systems. I am skilled in using various software and hardware related to electronics, such as ETAP, microcontrollers, Proteus, Autodesk Inventor, and others.

Work Experiences

PT. Energi Terbarukan Internasional - Surabaya, Indonesia

Feb 2025 - Jun 2025

Internship

A specialized EPC contractor in solar photovoltaic systems, providing comprehensive engineering, procurement, installation, and commissioning services for reliable and efficient renewable energy solutions

- (2025) Hybrid solar PV installation for a pet shop in Surabaya
- (2025) Hybrid solar PV installation for a tennis court in a premium residential area in Surabaya
- (2025) Feasibility design and analysis of a solar panel system for an industrial facility in Surabaya

PT. King Manufacture - Sukoharjo, Indonesia

Oct 2021 - Mar 2022

Internship

PT King Manufacture is a company engaged in automation, tool making, manufacturing, and plastic injection.

- (2021) Electrical and mechanical repairs on a 3-axis CNC milling machine to ensure optimal operation and accuracy
- (2022) Design and implementation of the electrical wiring installation for an industrial automation machine in the form of a wooden carrier train, ensuring efficient and reliable operation.
- (2022) Implementation of electrical safety emergency systems on a sawmill conveyor for wood cutting machines to enhance operational safety and meet industrial safety standards

PT. Alfian Mechatronics Innovation - Malang, Indonesia

Jul 2020 - Dec 2020

Internship

PT. Alfian Mechatronics Innovation is one of the companies engaged in contractors, suppliers, consultants, and industrial electrical automation engineering courses.

- (2020) Installation of Hybrid Solar Panel Systems (Off-Grid and On-Grid) for Residential Housing: Designed and implemented hybrid solar energy solutions to optimize energy savings and reliability
- (2020) Development of a solar panel trainer with pneumatic actuators to simulate solar tracking systems for educational and training purposes
- (2020) Implementation of 3000W Solar Panel-Powered Submersible Pump Water Well Construction
- (2020) Installation of Public street lighting at P4TK Batu

Education Level

State University of Malang - Malang, Indonesia

Undergraduate Electrical Engineering, 3.82/4.00

- (2023) Gold and Best Winner Infographic Award at ARAU International Creativity Exhibition
- (2023) Gold Winner at Syncounlogy: Service and Learning Product Competition
- (2023) Installation a Solar Power Plant (PLTS) to supply electricity for CCTV and Public Street Lighting (PJU) in Kedung Kandang Subdistrict, Malang City.
- (2024) Won 1st place in an essay competition organized by UKM Einstein, Faculty of Engineering, University of Jember
- (2024) Won 2nd place in The Abisatya Bussines Idea Development Competition 2024
- (2024) Part of the winning team for the Entrepreneurship Grant Competition for Students of the Department of Electrical and Informatics Engineering, Faculty of Engineering, Universitas Negeri Malang.

State Vocational High School 1 Purwosari - Pasuruan, Indonesia

Jul 2018 - Jun 2022

High School Diploma in Mechatronics, 86.35/100.00

- (2018) Training on 1-Phase Electrical Installation for Home Lighting: Completed training on the installation of 1-phase electrical systems, with a focus on residential lighting circuits, safety standards, and troubleshooting.
- (2019) Project Leader for Automatic Hand Dryer: Led the design and development of an automatic hand dryer system using sensors and electromechanical components, ensuring effective functionality and user safety
- (2020) Design of Electropneumatic Cascade Circuit: Designed and implemented electropneumatic cascade control systems for automation processes, ensuring efficient sequencing and functionality
- (2020) Development of an Automatic Rubik's Cube Solver: Designed and built an automatic Rubik's cube solver using a microcontroller-

based system, integrating mechanical arms and algorithms to solve the cube efficiently.

- (2021) Installation of 3-Phase Direct on Line (DoL) Motor Circuit: Installed and commissioned DoL motor starters, ensuring reliable and straightforward motor operation
- (2021) Installation of 3-Phase Forward-Reverse Motor Circuit: Installed and configured forward-reverse control for 3-phase motors, ensuring safe and efficient operation
- (2021) Installation of Star-Delta Motor Circuit: Performed installation and testing of star-delta motor starters for industrial applications
- (2021) PLC Trainer design as an implementation of industrial work
- (2021) Programming and Installation a Sorting Machine System as Competency Assessment

Organisational Experience

IPESG (TEAM A) - Malang, Indonesia

Oct 2023 - Present

Research Assistant

IPESG is a research and development team related to the field of Intelligent Power Electronics and Smart Grid at the State University of Malang, chaired by Aripriharta, S.T., M.T., Ph.D.

- (2023) Participation as an exhibitor at the Research and Community Service Product Exhibition by LPPM UM 2023, led by Aripriharta, S.T., M.T., Ph.D.
- (2024) Implementation of a Bottle Sterilization System with IoT-App Based Digitalized Production to support the development of MSMEs
- (2024) Design and development of an automatic grass chopper machine powered by solar panels as an effective solution for cattle feed management
- (2024) Development of a SCADA IoT Edukit for Collaborative Factory Automation Practices Among Engineering Students Using the VAK Learning Method
- (2024) DRTPM Research Project: Contribution to the Development of a Smart Picohydro Low Head System Using the QHBM Pressure Injection Method for Sustainable Green Economy in Rural Areas
- (2025) Pilot project on picohydro technology as an initiative to penetrate the renewable energy market through a startup

Skills, Achievements & Other Experience

- **Awards** 🏆 (2018): Member of the Flag Hoisting Troop (Paskibra) for the 73rd Independence Day of the Republic of Indonesia in Purwosari Subdistrict
- **Publications** 📄 (2024): Aripriharta, Imam Saputra, Satria Adiguna, Muhammad Afran Habibi, Arif Nur Afandi, Aditya Wahyu Setiawan, Muhammad Cahyo Bagaskoro. Design of Resistive Power Control Based on Microcontroller with TRIAC. Vol 7, No 1 (2024).
- **Projects** 🛠️ (2024): Installation of a Solar-Powered Water Pump for Rice Field Irrigation in Bantur Subdistrict
- **Copyrights** 📄 (2024): Fingerprint Flora
- **Achievements** 🏆 (2024): Speaker at the National Seminar on Community Service (Seminar Nasional Pengabdian Kepada Masyarakat) with the theme "Sinergi Perguruan Tinggi dan Masyarakat untuk Pembangunan Berkelanjutan"
- **Achievements** 📄 (2025): Technical manual book for the DRAGON-32 ESP32-based distance relay training system, designed to support laboratory learning and practical understanding of power system protection. The manual covers system architecture, hardware specifications, protection algorithms (impedance, reactance, and MHO relay), standard operating procedures, and fault simulation scenarios across multiple protection zones. Contributed to system design documentation, protection logic explanation, and development of structured practical modules for engineering education.